LECTURE 7

Welcome to lecture 7.

BLOOM'S DOMAINS OF LEARNING AND TAXONOMY OF EDUCATIONAL/LEARNING OBJECTIVES

TOPIC OBJECTIVES

After this lecture, the learner should be able to:

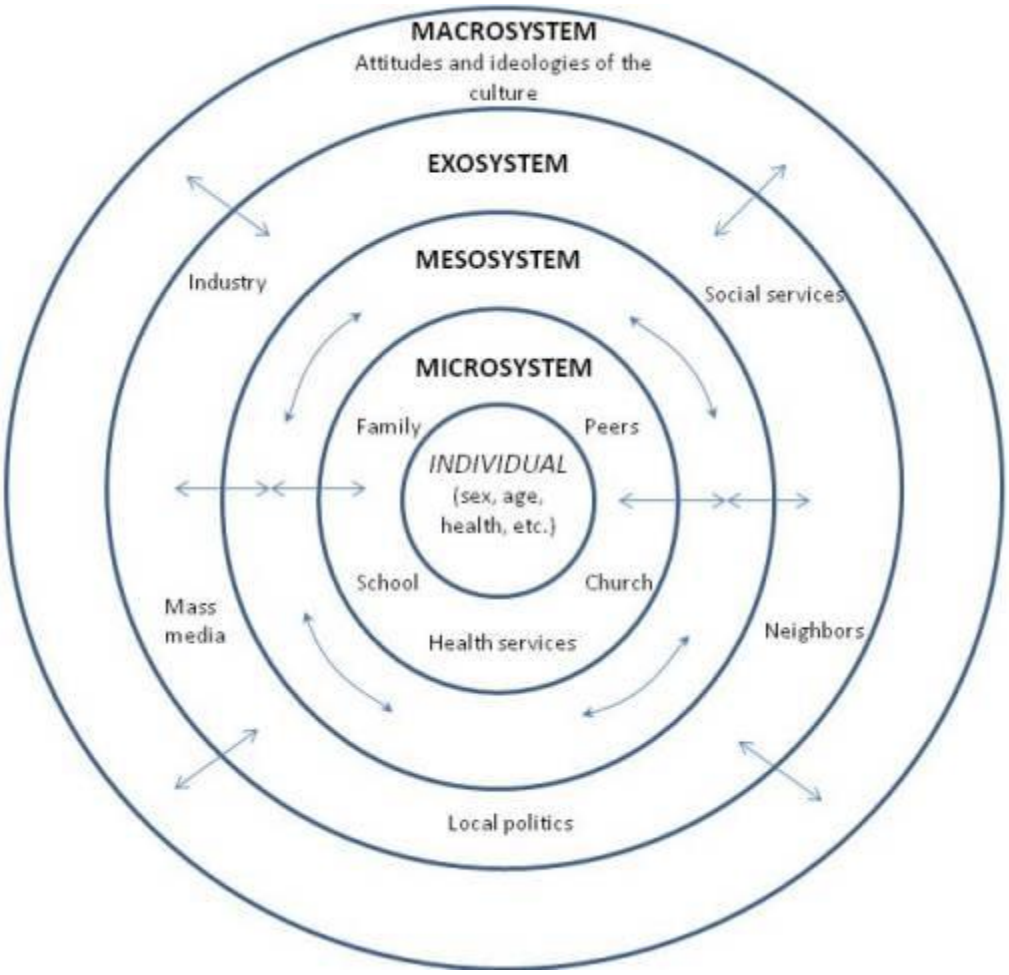
- Explain the term taxonomy.
- Distinguish between and discuss three types of Bloom's domains of learning.
- Explain the six categories of cognitive domains.

Taxonomy means classification. Taxonomies of educational objectives serve as sources for deriving instructional objectives and classifying learning behaviours. The taxonomies are classified into three domains of learning, namely:

- The ***Cognitive domain*** which deals with intellectual development (Bloom, 1956);
- The ***Affective Domain*** which deals with the development of feelings, values and attitudes. Issues of integrity and corruption are within the affective domain (Krathwohl, 1964); and
- The ***Psychomotor domain*** which deals with motor / practical / manual skills development (Simpson, 1972).

From the ***Ecology of Human Development*** perspective, the diagram below summarizes the influences of the children's learning of values and attitudes.

- Child vs. John Locke concept of ***tabula rasa*** (blank slate)
- Society plants hate virus in children
- From the diagram, where do you put Kenya's leaders hate speeches?



Of the three, the ***Cognitive domain*** is often the most emphasized especially in formal school settings but the other two are also used whenever the subject to be learnt can be used to teach skills, attitudes, values and feelings. Here, below, are some action verbs which demonstrate observable behaviors under the three domains of learning.

For cognitive domain: copy, classify, apply, analyse, define, identify, compare

For affective domain: accept, discuss, dispute, join, defend,

For *psychomotor domain*: adapt, adjust, assemble, build, operate

SETTING EDUCATION OBJECTIVES BASED ON COGNITIVE DOMAINS

Blooms Six Categories of Cognitive Domains

Bloom's taxonomy contains six categories of cognitive skills ranging from lower-order skills that require less cognitive processing to higher-order skills that require deeper learning and a greater degree of cognitive processing (see Figure 1 below).

Bloom and his colleagues categorized cognitive domain into 6 categories from simple to complex or from factual to conceptual. Whenever you are setting instructional objectives that are intended to measure cognitive domain you apply the following Blooms 6 categories of cognitive domain.

1. **Knowledge** (recalling information). This is the lowest level of objectives e.g. memorizing mathematics facts or formula, scientific principles. Knowledge objectives require students to reproduce facts e.g. formula for calculating area of a triangle.
2. **Comprehension** (translating, interpreting). Comprehension, objectives require that students show an understanding of information as well as the ability to use it. Examples include: interpreting the meaning of a diagram, graph or parable.
3. **Application** (using principles or abstraction to solve novel or real life problem). Application objectives require that students apply theoretical knowledge or information to solve a real-life problem. (KALRO research activities fall here)
4. **Analysis** (breaking brown complex information or ideas into simpler parts to understand how the parts relate or are organized). Examples of analysis objectives include: contracting schooling in Kenya with education in Japan or Tanzania.
5. **Synthesis** (creation of something that did not exist before). Synthesis objectives involve using skills to create completely new products. Examples include designing as science experiment to solve a problem. Further example, coming up with an innovation. Converting ideas into a product. Coming up with an innovation eg. Converting ideas into a product
6. **Evaluation** (judging something against a given standard). Evaluation objectives require the students to make value judgments against some criterion or standard.

Knowledge } Testing lower level abilities

Comprehension

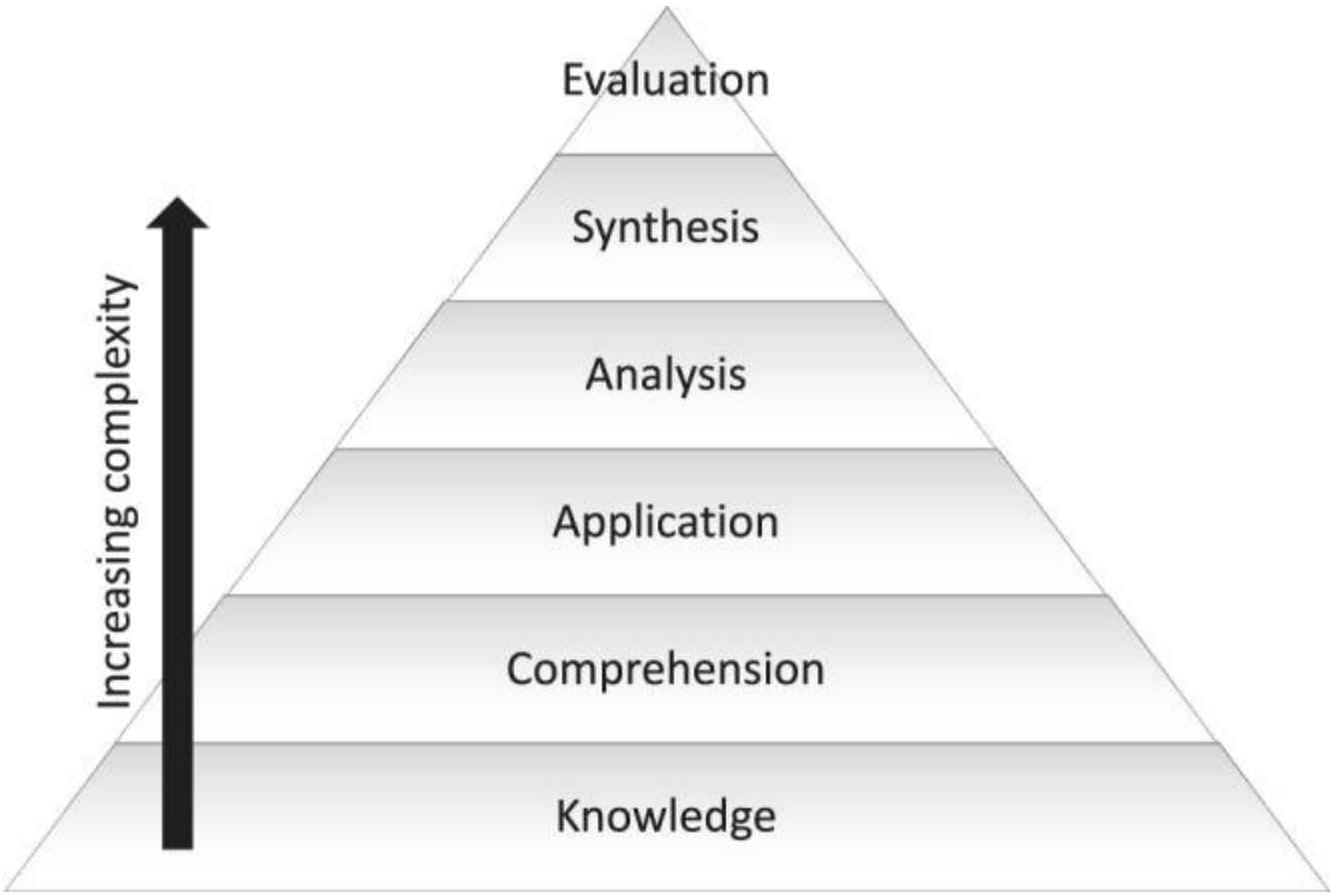
Application } Testing medium level abilities

Analysis

Synthesis } Testing higher level abilities

Evaluation

In testing try to balance test items that measure lower level abilities and those that measure higher level abilities.



Bloom's Six Categories of Cognitive Domain

ACTION VERBS TO USE FOR EACH COGNITIVE DOMAIN

Knowledge	Comprehension	Application	Analysis	Synthesis	Evaluation

copy	classify	apply	analyse	arrange	appraise
define	describe	change	appraise	assemble	argue
duplicate	explain	demonstrate	calculate	collect	assess
label	express	discover	categorize	combine	attach
list	identify	dramatize	compare	compose	choose
match	indicate	draw	contrast	create	compare
memorize	interpret	employ	criticize	design	conclude
name	locate	extend	diagram	devise	defend
order	outline	illustrate	differentiate	formulate	estimate
quote	relate	manipulate	distinguish	manage	evaluate
recognize	report	modify	examine	manipulate	judge
recall	respond	operate	experiment	modify	justify
record	restate	perform	explain	organize	predict
repeat	review	prepare	illustrate	originate	rate
reproduce	rewrite	produce	predict	plan	score
tell	select	show	question	propose	select
underline	translate	solve	test		support
	set up	use			value

REVISED BLOOM'S TAXONOMY [NOT FOR DISCUSSION]

Based on findings of cognitive science following the original publication, a later revision of the taxonomy changes the nomenclature and order of the cognitive processes in the original version. In this later version, the levels are ***remember, understand, apply, analyze, evaluate,*** **and** ***create.*** This reorganization places the skill of synthesis rather than evaluation at the highest level of the hierarchy. Furthermore, this revision adds a new dimension across all six cognitive processes. It specifies the four types of knowledge that might be addressed by a learning activity: *factual* (terminology and discrete facts)*; conceptual* (categories, theories, principles, and models)*; procedural* (knowledge of a technique, process, or methodology); and *metacognitive* (including self-assessment ability and knowledge of various learning skills and techniques).

ACTION VERBS TO USE FOR AFFECTIVE DOMAIN

accept	discuss
accept	display
attempt	dispute
challenge	follow
change	form
commend	initiate
comply	integrate
conform	join
defend	judge

ACTION VERBS TO USE FOR PSYCHOMOTOR DOMAIN

Adapt	Duplicate	Move	Select
Adjust	Fix	Operate	Service
Assemble	Generate	Perform	Set up
Bend	Grasp	Pickup	Shorten
Build	Handle	Point to	Show
Calibrate	Hear	Practice	Slide
Close	Identify	Press	Sort
Combine	Illustrate	Pull	Stretch
Construct	Load	Push	Touch
Copy	Locate	Remove	Transport
Design	Loosen	Repair	Write
Diagram	Manipulate	Replace	
Disconnect	Measure	Rotate	
Draw	Modify	See	

SELF ASSESSMENT EXERCISE TO BE DISCUSSED IN CLASS

Underline the verb phrases which best demonstrates observable behaviors:

Know	State	Describe
Give examples of	Understand	Really know
Fully understand	Suggest reasons why	Explain

Evaluate	Be familiar with	Become acquired with
Pick out	Distinguish between	Have a good grasp of
Appreciate	Analyse	Carry out
Summarize	Compare	Acquire a feeling for
Summarize	Learn the basics of	Realize the significance of
Believe in	Demonstrate	Show diagrammatically

SAMPLES OF AIMS AND LEARNING OBJECTIVES

Learning objectives must be stated in **action verbs.**

Sample 1: Course in Literature in English

Aims: To develop skills in literary appreciation through reading novels.

Learning Objectives: After reading Chinua Achebe's novel, Things Fall Apart, a learner should be able to:

- Explain the title of the novel
- Compare the character of Okonkwo with that of his father
- Give his/her opinion about Okonkwo's killing of Ikemefuna.

Sample 2: Course in Business Management

Aims: To help the non-financial manager use financial information in planning, controlling and making decisions.

Learning objectives: At the end of this course, a learner should be able to:

- Construct a cash flow forecast.
- Use a profit and loss account.
- Interpret a balance sheet.

- Describe ways of classifying expenditure.

LECTURE 8

Welcome to lecture 8.

INSTRUCTIONAL/LEARNING OBJECTIVES AND LEARNING OUTCOMES

TOPIC OBJECTIVES

After this lecture, the learner should be able to:

- Explain the concept of:
- Learning objectives.
- Learning outcomes.
- Explain why action verbs and SMART principle are necessary in stating learning objectives.
- Write your own learning objective.

Whenever you plan for class instruction, you must set instructional learning objectives of the course.

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What are **learning objectives**? Objectives are specific statements of what is to be **accomplished and how well**, and are expressed in terms of action-verbs; quantifiable, measurable outcomes. Objectives address the questions:

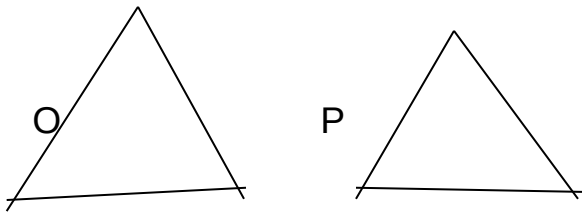
- What the course/lesson/programme can do or expects the learner to do at the end the course or training.
- What are **learning outcomes**? What the learners are able to do after exposure to a period of learning or training. Contextually demonstrated knowledge, skills and values after the lesson/course.

How do we measure the achievement of course objectives? By testing the learners.

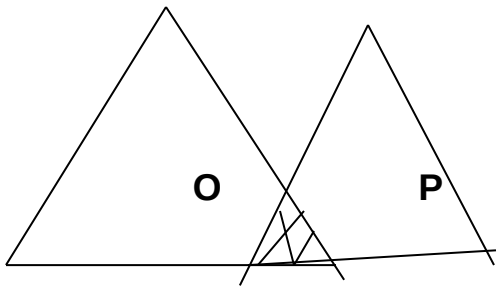
Objective *Performance*

After teaching the course you may have the following scenarios:

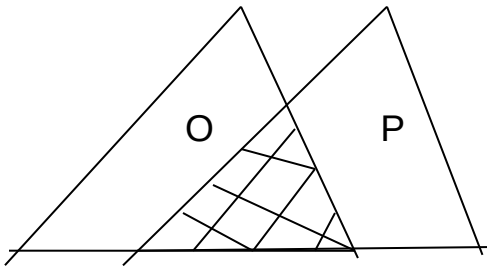
i. If the two triangles do not intersect, the course objectives were not met.



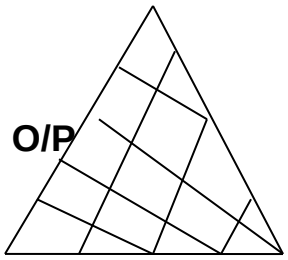
ii. The course objectives and performance may interact at 10% meaning 10% of the objectives were achieved.



iii. The course objectives and performance may intersect at 50% meaning that 50% of the objectives were achieved.



iv. The course objectives and performance may intersect 100% (in total congruence) meaning that 100% of the objectives were achieved.



1. What are the instructional/Learning objectives?

- Instructional objectives describe desirable outcomes – the skills, knowledge, abilities or attitudes – learners should possess or demonstrate after they complete the course or training. Statements of intend. What the course/program can do. In short, they are tools for describing student outcomes.
- Learning objectives focus on student performance. Must be measurable, seen or heard; expressed in action verbs and must be SMART. A learning objective is a statement that specifies in behavioural (measurable – seen or heard) terms what a learner **will be able to do (performance)** as a result of exposure to instruction.

That is, after completing the course / training. The number of learning objectives will determine the content of the subject matter to cover and exercises / activities to include. Too many objectives confuse the learner and too few, wouldn't communicate adequately what you intend for the learner. A rule of thumb should be to formulate as many learning objectives as would lead a learner to attain the knowledge, abilities and skills you intend to cover in a given unit / topic.

2. Importance of instructional objectives

- **Provides Direction to Learning**. Why are instructional objectives key to effective instructions? Instructional objectives provide both the teacher and the learner with direction for instruction. Instructional objectives focus *on learning, not teaching activity*. They state the instructional intend in terms of learning.
- **Old saying**: The importance of stating objectives has been succinctly expressed in the statement that if people do not know where they are going or what they are doing, they will be unable to find out whether they have arrived or what they have achieved.
- **Assist in Planning Instruction**: Instructional objectives are important for planning instruction in the following way:

Tell or inform the learners what is expected of them, and hence serve as guide for them. Objectives give direction to learning. Objectives help a teacher plan instruction, guide student learning and provide criteria for evaluating student outcomes.

- **Guide Teaching**: Involve the determination of the media, materials and procedure that is necessary to facilitate the learning rational sequencing of instruction.
- **Facilitate Evaluation**: Assist in determining the appropriate methods for evaluation. [Evaluating students' learning activities, instructions, students and curriculum]

3. Writing Instructional Objectives

In formulation of objectives, it will be helpful to ask such questions as:

(i). What will the student be able to do as the result of instructions? First, determine the learning outcomes for the lesson. That is, what should the learner be able to do when the lesson is over (outcome)?
(ii). How can they demonstrate achievement with pencil-and-paper behavior?
(iii) How to define objectives.

Instructional objectives **MUST** meet two major criteria:

(i) They must be stated in **ACTION VERBS.**
(ii) They must be **SMART.** SMART is acronym for:

S = Specific: Ensure that the objectives are specific. Specification of the end results of learning – the abilities which learners will have acquired when they have achieved the objectives e.g. will be able to add 1 + 1 or identify the characteristics of living things.

M = Measurable: Can the objectives you have set be measured? Objectives must be expressed in behavioural terms – indicating what the learner will be able to do at the end of the instruction.

A = Achievable: Are the objectives achievable or realistic?

R = Relevant: Are the objectives relevant to the topic or programme's purpose?

T – Time-bound: What is the time-line for achieving the set objectives? Eg. at the end of the course, training, semester, academic year?

Examples of poorly stated objectives

Course / instructional objectives provide the basis for test construction. That is, the content and structure of a test must be based upon the objectives of instruction. For example: Topic of osmosis. At the end of the lesson the learners should be able ***describe*** the process of osmosis.

Poorly stated objectives cannot be measured or fail to provide the basis for measurement of pupil achievement. Examples of poorly stated objectives are:

At the end of the course, learners must be able to

1. "Know the important facts about osmosis".
    - How can we determine when a learner "knows?"
    - What must a pupil be able to do in order to demonstrate what he knows?
2. To "understand Newton's Law of Motion"
    - How can it be determined when a learner "understands" or the depth and breadth of his understanding?
    - What must the pupils be able to do in order to demonstrate his understanding?
3. To "become familiar with the services rendered by the Teachers Service Commission".
    - How "familiar" should a student be?
    - How can we measure "familiarity" with ideas or information?

4. Classification of Instructional Objectives

Teachers should classify objectives because the type of objectives dictates the selection of instructional methods, materials, media and evaluation to be used in the lesson. Objectives may be classified according to the primary learning outcomes that take place. These learning outcomes typically are classified into three domains / categories as discussed earlier in the course:

- Cognitive
- Affective
- Psychomotor

LEARNING OUTCOMES

- Learning outcomes are what students are able to do after a period of learning or lesson.
- Contextually demonstrated knowledge, skills and values after the lesson/course.
- Change in behavior that an individual demonstrates after instructional period.
- Ability of a learner to demonstrate knowledge, skills, attitudes that he /she has acquired after undergoing a learning process.